COMP4020 Project Proposal

NewsDigest: News Collection and Automatic Summarisation

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GitHub Repository: https://github.com/thaibahung/NLP-Project

1 Introduction

The growth of digital news has made it increasingly difficult for readers to keep up with current events efficiently. News websites publish a large number of articles every day, and many articles are lengthy or repetitive across different sources. As a result, readers often spend significant time identifying the most relevant information. This project proposes an NLP-based system that collects online news articles and automatically generates concise summaries. The main Natural Language Processing task is **text summarisation**, and the application domain is **digital news media**. The project is designed as a practical university-level system that combines data collection, text preprocessing, summarisation, and evaluation in one complete pipeline.

2 Project Objectives

This project has three objectives. First, it aims to build a simple and reliable pipeline for collecting and cleaning news articles from publicly accessible sources. Second, it aims to apply summarisation methods to reduce article length while preserving the key information. Third, it aims to produce a prototype that demonstrates how NLP can support faster and more efficient news reading. The project addresses the question: *How effectively can an NLP system summarise newspaper articles in a way that is concise, readable, and informative?*

3 Proposed Method

The project will be implemented in four stages. First, the team will collect a set of news articles and organise them into a structured dataset. Second, the text will be preprocessed by removing HTML tags, duplicate content, punctuation noise, and other irrelevant webpage elements. Third, the team will experiment with two levels of summarisation methods: a simple baseline method and a more advanced transformer-based model. For the advanced approach, the project plans to use BART, which has shown strong performance in text generation and summarisation tasks. Finally, the generated summaries will be evaluated using ROUGE scores and a short qualitative review of readability and informativeness.

4 Dataset

The main dataset for this project will be the **CNN/DailyMail** news summarisation dataset, which contains English news articles paired with human-written highlights. This dataset is suitable because it is widely used in summarisation research and reflects realistic news-writing style. As a secondary dataset, the project will use **XSum**, which consists of BBC news articles paired with highly concise one-sentence summaries. Using both datasets allows the team to compare summarisation performance across different summary styles. In addition, the team will collect a small number of recent articles through web scraping for demonstration purposes in the final prototype. The dataset therefore includes both benchmark data and real-world input. Expected challenges include noisy webpage text, duplicated articles across sources, and variation in article length and writing style.

5 Expected Final Product

The final product will be a prototype news summarisation system. A user will be able to input or select a news article and receive a short automatically generated summary. The system is intended to improve the manual process of reading full-length news articles by reducing time and effort while preserving the main ideas. In addition to the prototype, the project will provide a comparison of methods, a short evaluation of summary quality, and a discussion of the system's strengths and limitations.

6 Team Responsibilities

- **Member 1:** Data collection, web scraping, and dataset organisation.
- **Member 2:** Text cleaning, preprocessing, and exploratory analysis.
- **Member 3:** Summarisation modelling and experimentation.
- **Member 4:** Evaluation, visualisation, system integration, and final report preparation.

The responsibilities are balanced across the full NLP pipeline and ensure that each team member contributes to a specific stage of the project. Overall, the project is feasible for a team of four and is suitable for a university-level NLP course because it combines a clear real-world problem, established datasets, and a manageable technical scope.

References

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